Modern Data Engineering Architecture

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SECP3843 : Special Topic In Data Engineering
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Executive Summary

This report examines Modern Data Engineering Architecture and its role in enabling scalable, flexible and real-time data-driven systems in today's AI-driven digital environment. The paper examines how data architecture has evolved over the years from conventional data warehouses to new paradigms like data lakes and lakehouse models, with the transition to cloud-native and distributed systems. Major findings include NoSQL databases to ensure horizontal scalability and high availability, cloud computing to provide elastic resource usage and ELT-based pipelines to enhance the efficiency of processing raw data by storing the data before transformation. Moreover, AI/ML and MLOps integration can improve automation, model lifecycle management, and efficiency in data engineering processes.

The report further identifies a modern multi-layer architecture consisting of data ingestion, storage, processing, serving and orchestration layers. Both batch and real-time analytics are supported in this architecture, hence suitable for applications such as e-commerce personalization and ride-hailing platforms.

Despite the advantages of modern data engineering systems, challenges remain such as high complexity, cost management and data governance issues. Security and data quality also remain as a critical concerns in the distributed environments.

As a solution, adopting cloud-based or hybrid infrastructures is recommended. Real-time processing tools such as Apache Kafka and Apache Spark should be utilised and workflow orchestration implemented using platforms like Apache Airflow. Strong governance frameworks, including encryption, access control and continuous monitoring are also essential for maintaining system reliability and compliance.

In conclusion, modern data engineering architecture is fundamental for supporting intelligent, data-driven enterprises. Its continued evolution driven by AI integration, data mesh concepts, serverless computing and lakehouse models, will further enhance its capability to deliver timely, scalable and actionable insights for competitive advantage.

1.0 INTRODUCTION

In the current era of intelligence connectivity, data is no longer a passive byproduct from business operations, or rather, it is now the primary strategic resource that drives enterprise systems. Modern Data Engineering Architecture refers to discipline that guides the proper construction of complex data processing systems and addresses challenges in terms of data velocity, variety and volume.

Unlike traditional frameworks, modern architecture emphasizes scalability, flexibility, and real-time data processing by leveraging cloud-native technologies, separating storage from compute and adopting ELT (Extract-Load-Transform) approaches that allow raw data to be stored and processed more efficiently. It also supports diverse data types and enables faster or real time insights through distributed and parallel processing systems.

This report investigates the key components, principles and technologies of Modern Data Engineering Architecture, as well as how they overcome the limitations of legacy systems to support data-driven and intelligent decision-making.

2.0 LITERATURE REVIEW

The data management system has evolved over the years due to the growth in data amount across industries. Organisations have shifted from traditional relational database technologies towards NoSQL solutions to manage big data. (Ali et al., 2023) To accommodate the expanding scope and complex modern data application, database technologies have undergone substantial changes. (Khan et al., 2023)

Data architecture has progressed through multiple stages, from traditional data warehouse, data lakes to data lakehouse architectures. Traditional data warehouses store structured filtered data, which are not able to handle efficiently even after the advancement of storage capacity and high-speed processors due to their rigid schema structure. (Harby & Zulkernine, 2022) The traditional schema design limits scalability and flexibility when handling distributed data modifications across clusters. These constraints have driven the need for more adaptable and scalable architectural approaches. (Khan et al., 2023)

In response to these challenges, cloud-native database systems have become a critical component of modern data engineering architecture. This was due to the increasing demand for scalability, elasticity and on-demand usage by modern applications. Cloud-native design architecture such as shared storage enables the decoupling of compute and storage which enhances system elasticity and thus allows efficient resource utilization. (Li, 2019) Consequently, cloud computing has become a foundational platform for digital transformation initiatives due to its ability to support scalable and flexible data infrastructures. (Nagabhyru, 2023)

As data volumes and analytical demand grows exponentially, it becomes impractical to rely on manual coding alone for data processing and pipeline management because of the limited skillset one has. (Nagabhyru, 2023) This has led to the development of methodologies and tools aimed at improving efficiency and reducing the operational burden associated with data engineering tasks. Machine Learning Operations (MLOps) addresses the challenges of automating and operationalizing machine learning workflows. By introducing structured processes for automating the lifecycle of machine learning models, MLOps facilitates the integration of development and deployment practices within data-driven systems. (Dominik et al., 2023) In addition, AI-driven data engineering approaches further enhance the ability to manage increasing data complexity by aligning data workflows with business objectives, while simultaneously reducing the need for extensive manual intervention. (Nagabhyru, 2023)

There are several challenges that remain in the implementation of modern data engineering architectures. Integrating machine learning capabilities directly into database systems are difficult due to the computational cost associated with training and inference processes. (Li, 2019) Additionally, ensuring the availability and quality of data pose a challenge in monolithic architectures such as traditional data lake systems, where discovering relevant and high-quality data can be time-consuming. (Araújo et al., 2021) Another critical issue is the shortage of skilled professionals, including data architects, data engineers, machine learning engineers and DevOps specialists, which further complicates the development and maintenance of complex data systems. (Dominik et al., 2023)

In response to these limitations, new architectural paradigms continue to emerge. The data lakehouse model has been proposed as a hybrid solution that combines the flexible and scalable storage capabilities of data lakes with the structured and optimized data management features of data warehouses. (Harby & Zulkernine, 2022) This approach aims to address the limitations of both traditional systems while supporting a broader range of analytical workloads. Furthermore, with the increasing adoption of cloud computing, cloud-native database systems are expected to play an increasingly important role in the future of data engineering (Li, 2019), providing the scalability, flexibility, and performance required to support next-generation data-driven applications.

3.0 ANALYSIS AND DISCUSSION

Modern data engineering architecture involves collecting, processing, storing and using data efficiently. In the past, most systems used batch processing, which meant that data was processed only at certain times. However, modern systems are more advanced and can process data in real time, enabling organizations to make faster and more accurate decisions (Reis & Housley, 2022).

3.1 HOW THE TECHNOLOGY WORKS

Multi-layered data architecture is common nowadays. The first layer is the data ingestion layer, which collects data from multiple sources like databases, websites, mobile apps and sensors. Data can be ingested in batches or continuously.

Second, is the data storage layer, where the data is stored in systems such as data lakes or data warehouses. A data warehouse stores structured data for analysis, whereas a data lake stores raw data in its original form (Kleppmann, 2017).

Then, the data processing layer cleans, transforms and arranges the data, making it fit-for-purpose for efficient analyses. This processing can be done in batches or real-time, depending on the system design.

The data serving layer provides the user with processed data through dashboards, reports or machine learning models.

Finally, orchestration and monitoring manage workflow and make sure that processes run smoothly and efficiently (Evans, 2023).

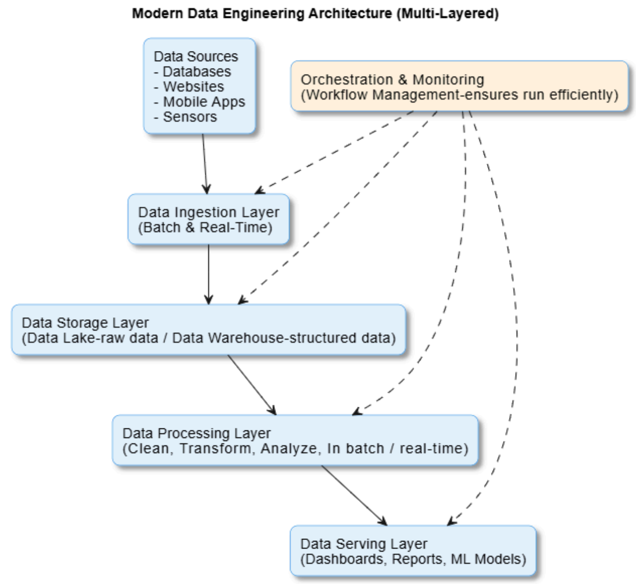

Diagram 3.1.1 shown the multi-layer of modern data engineering architecture

3.2 CASE EXAMPLES / APPLICATIONS

One example of modern data engineering architectures is e-commerce platforms. These systems collect user data including clicks, searches and purchases and quickly process it to provide personalized recommendations and improve user experience.

Applications for ride-hailing are another example, which use real-time data to match drivers and passengers, estimate trip times and calculate fares. These services wouldn't operate effectively without modern data architecture.

3.3 ADVANTAGES AND DISADVANTAGES

Modern data engineering is fast, scalable, flexible and reduces hardware needs (Reis & Housley, 2022). However it can be complex, costly and hard to secure (Kleppmann, 2017). The table below shows the advantages and disadvantages.

Table 3.3.1 Table shown advantages and disadvantages of modern data engineering architecture

Advantages	Disadvantages
Highly scalable, can handle large volumes of data	Complex and difficult to manage, especially for beginners
Allows real-time processing to speed up decision-making	Can become costly if cloud resources are not properly controlled
Flexible and can manage different types of data	Maintaining data quality and ensuring security can be challenging

3.4 CHALLENGES AND FUTURE TRENDS

A major issue is ensuring data quality and consistency, as data is collected from different sources. Maintaining data security and privacy is another problem, especially when handling sensitive data. Since delays can affect system performance, latency is also an issue in real-time systems. In addition, there aren't enough experienced professionals in this field.

In the future, many trends are developing. One is data mesh, which divides up data ownership among groups. The increasing use of real-time analytics is another trend. In order to obtain more deep insights, organizations are also integrating data engineering with artificial intelligence. Additionally, serverless technology is growing in popularity and eliminating the need for infrastructure management. Lastly, lakehouse architecture is developing, combining data warehouse and lake features into one system (Evans, 2023).

4.0 RECOMMENDATIONS

According to the analysis of modern data engineering architecture, organizations have to focus on developing systems that find a balance between security, scalability and flexibility. It is recommended to start with cloud-based and hybrid storage solutions. Cloud services, like Amazon Web Services, allow companies for growing processing and storage capacity as needed while reducing the expense of physical infrastructure (Amazon Web Services, 2024). Combining cloud and on-premise storage can also provide better control over sensitive data.

Besides, organisations also should use tools such as Apache Spark and Kafka to create real-time data processing pipelines. In dynamic environments like e-commerce, logistics and ride-hailing services, real-time analytics allow quicker decision-making and enhance responsiveness. To maintain consistency and reliability in data pipelines, it's also important to automate workflows using orchestration technologies like Apache Airflow.

Lastly, it is important to prioritise data security and governance. Clear policies for encryption, access control and data quality must be established by organisations. Regular monitoring and audits will help in maintaining regulatory compliance and preventing data breaches (Kleppmann, 2017).

Organisations may create data systems that are reliable, economical and able to provide timely insights to support business decisions by following these recommendations.

5.0 CONCLUSION

Modern data engineering architecture is important to an organisation's ability to effectively handle and analyze large volumes of data. It is suitable for today’s data-driven applications because it enables real-time processing, scalability and flexibility. However technology also brings with it challenges such as data security, cost control and system complexity.

Overall, for organisations to gain the greatest rewards, they must carefully plan and administer their data systems. Future developments in data engineering approaches will be expected to include real-time analytics, data mesh and AI integration. Organisations will have to continuously learn and adapt in order to remain competitive and make better data-driven decisions.

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APPENDIX / LOGBOOK

Date	Search Keywords	AI Prompts Used	Progress	Member
1 April 2026	modern data engineering architecture, data engineering architecture design, big data architecture, modern systems, data pipeline architecture, data lakehouse architecture, cloud-native data architecture	Suggest topic keywords for finding relevant research paper about modern data engineering architecture	Did research on relevant topics to understand modern data engineering architecture	Tan Yi Ya

4 April 2026	Modern data engineering architecture, data pipeline, cloud data	None	Searched Scopus and Google Scholar for articles.	Teh Ru Qian
4 April 2026	Challenges and limitation of legacy system, cloud native technology	Briefly explain the limitations of legacy data architecture and the turning point for modern architecture	Wrote the introduction section and outlined the overall direction of the report.	Tan Yi Ya
5 April 2026	Data architecture layers, real time application	Give simple examples of data engineering applications	Read and compared sources. Wrote the explanation for each layer architecture.Explained how real-time data used in real world situations.	Teh Ru Qian
6 April 2026	Advantages and disadvantages	Help summarize advantages and disadvantages.	Completed advantages and disadvantages in table form.	Teh Ru Qian
7 April 2026	Cloud architecture AWS, data engineering best practices	Give recommendations for organizations using data architecture	Explained suggestions like cloud adoption, real-time processing and data governance.	Teh Ru Qian
8 April 2026	Future trends, data mesh, lake house	Explain data mesh in simple words.	Wrote challenges and future trends.	Teh Ru Qian
9 April 2026	(Referred to source)	Based on the topic provided, suggest outline and flow for the literature review section.	Verified outline and extracted information from the source. Paraphrased and cited every work.	Tan Yi Ya
9 April 2026	APA citation format	None	Finalized conclusion,checked grammar and	Teh Ru Qian

			formatted references.	